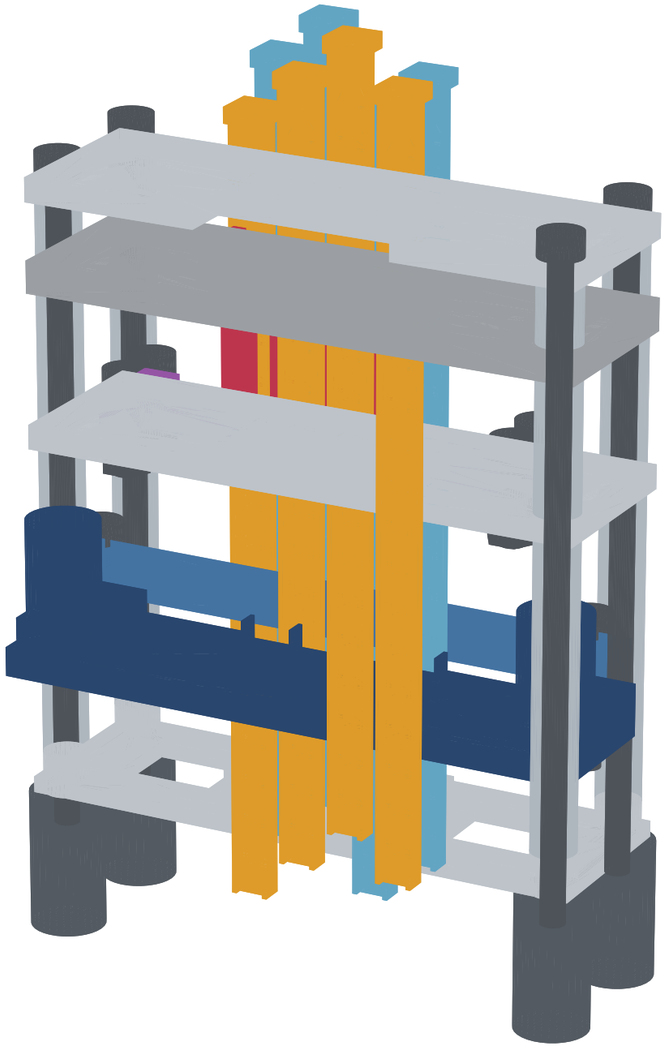
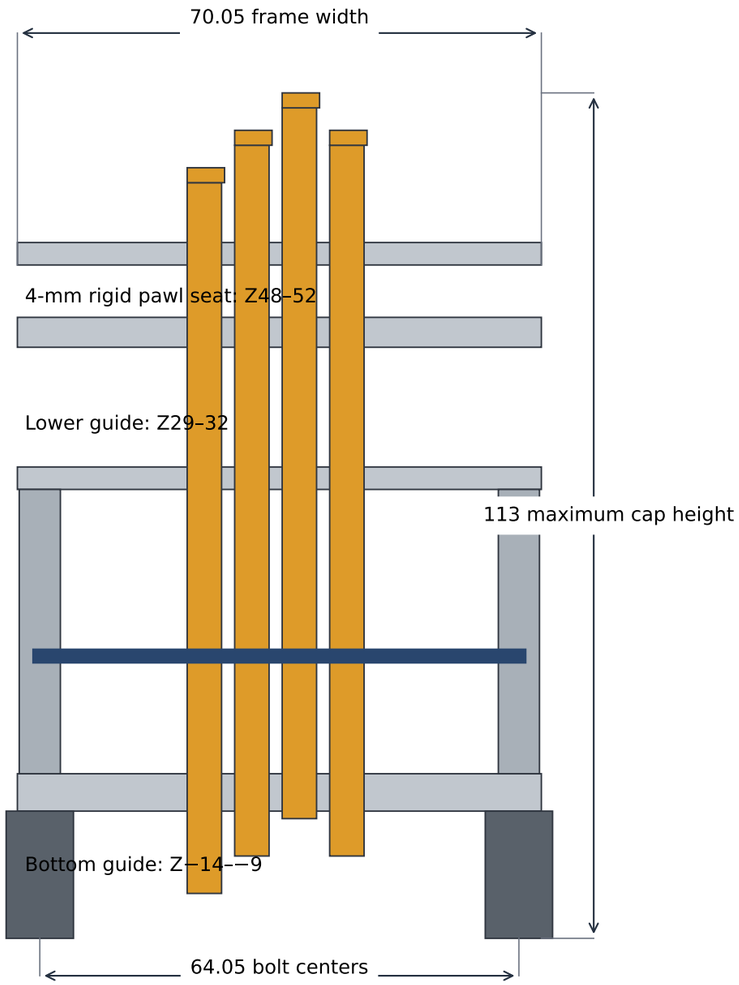
Four active racks + four untouched witnesses

6.35-mm pitch • 0–10-mm permitted terrain heights • signed 5-mm pulses • manual selection and load handoff



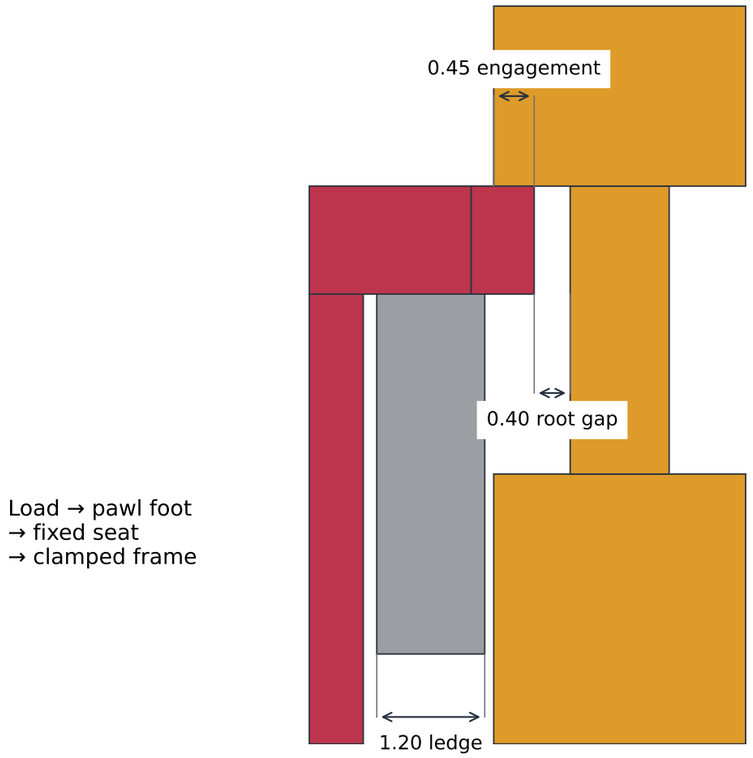
Orange: active racks. Blue: witnesses. Red: stationary ground pawls.
Green: moving captured dogs. Purple: clamped comb roots. Grey plates carry load.



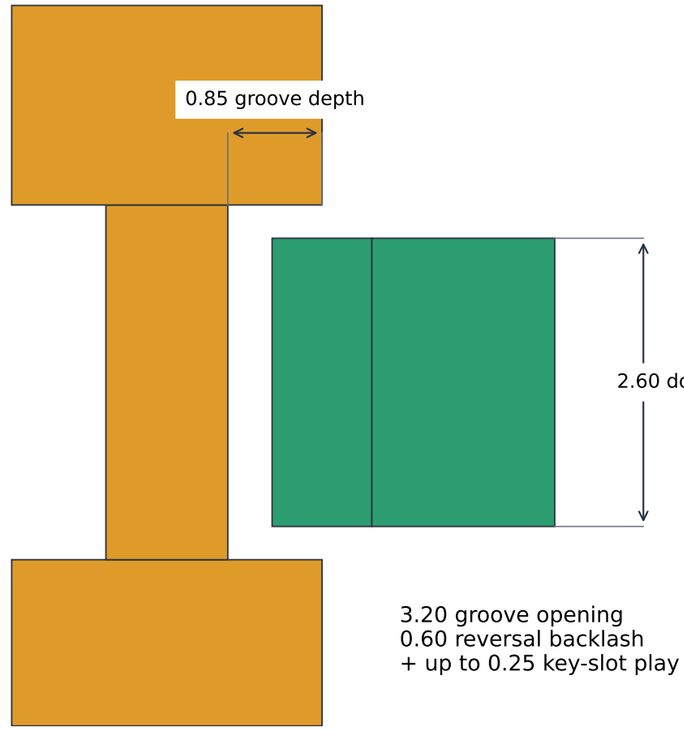
Frame stack, bottom to top:
5 guide + 38 sleeves + 3 guide + 16 sleeves + 4 seat + 7 sleeves + 3 guide = 76 mm.
M3×80 heads above; 2.4-mm nuts below; nominal 1.6-mm thread protrusion.
Four 17-mm gravity foot cups clear nuts and keep pin tips 6 mm above bench.

Nominal contact geometry and manual transaction

Ground support: middle of groove, X=0



Captured dog: both groove flanks are functional



Rail and guidance dimensions

Pin 4.6 × 2.8; guide aperture 5.0 × 3.0; groove width 3.4; tooth width 2.4.
Continuous 0.6-mm X-edge lands preserve Y guidance through every rack phase.
Sleeves OD 5.5 / ID 3.2; carriage bore 5.9, effective bearing length 18.4.
Pin guides at Z−14…−9, Z29…32 and Z59…62. Cap 5×5×2; pin body 95 mm.
Carrier floor 3.6; lid 6.0; root-clamp underside relief 0.30.

Order is essential

1. Capture only selected dogs; lock both ON and OFF handles.
2. Raise gently until selected ground pawls are visibly unloaded.
3. Retract only those pawls and insert their grounded keepers.
4. Move until every target groove admits its ground pawl.
5. Restore all pawls; seat and retire each dog individually.

These sections omit continuous edge rails outside X=0. Clearance arithmetic is nominal.
Use observed pin/groove alignment for lowering; head displacement alone does not prove forced descent.
Inspect every pawl and keeper. Stop on binding, lost support, damaged fingers or unintended neighbor motion.